

Uncertainty-Aware Machine Learning for Solvent Recovery Flowsheet Evaluation

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Roadmap

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- Civilian service at the Institute for Ecopreneurship
- My role: use machine learning to find solvent separation strategies
- Previously
 - 2026–: Freelance musician and machine learning engineer
 - 2024–2026: BA Tonmeister at Zurich University of the Arts (ZHdK)
 - 2023–2024: MSc Machine Learning at University College London (UCL)
 - 2018–2022: BSc Computer Science at the University of Basel

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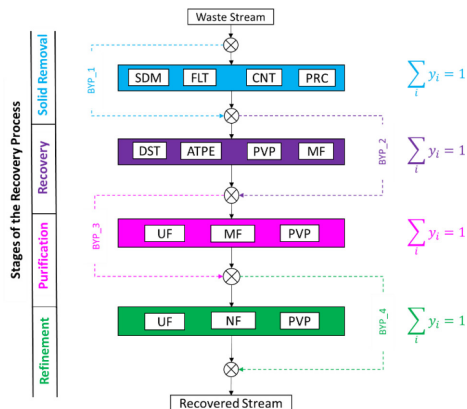
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- Chemical and pharmaceutical processes produce solvent-laden waste streams (Seyler et al., 2006).
- Recovering the solvent is often cheaper and greener than incineration – directly relevant to life cycle assessment.
- **But:** there are many possible ways (*flowsheets*) to recover a solvent. Which one is best?

The superstructure idea (Chea et al., 2020)

Enumerate *all* candidate recovery pathways in a single network of unit operations (sedimentation, distillation, pervaporation, ...) and let an optimiser pick the cost-optimal path.

The superstructure



- 4 stages: solids removal → recovery → purification → refinement
- Each stage: a handful of unit operations, or a bypass
- $4 \times 4 \times 4 \times 5 = 320$ distinct pathways

Figure 1: Superstructure inspired by Chea et al. (2020), reproduced from Aboagye et al. (2021).

What's missing from the optimiser?

The original approach solves a mathematical optimisation problem (MINLP) and returns *one* answer: the optimal path and its cost.

Two things a point estimate cannot tell you

- ❶ **Confidence.** How sure are we? The underlying cost model has a reported deviation of $\sim 30\%$.
- ❷ **Trade-offs.** How much worse is the second-best path? Maybe it is 1% more expensive but far more robust.

Goal of this project: a fast model that evaluates *any* pathway and tells us *how certain it is* about its own answer.

What is machine learning?

One-sentence definition

Instead of programming the rules by hand, we show a flexible mathematical function many examples of inputs and correct outputs, and let an algorithm adjust the function's parameters until its predictions match.

- The relevant paradigm is called *supervised learning*: we supervise the model with known input–output pairs.
- The goal is for the fitted function to *generalise* – it gives good answers for inputs it has never seen.
- Deep learning is the most prevalent and often the most powerful approach, using neural networks to approximate a function given data.

Neural networks

- A neural network is a chain of simple building blocks: stacked linear combinations passed through non-linear activation functions.
- Neural networks are capable of learning and reproducing highly complex relationships in high-dimensional data, from audio (e.g. Suno) to text (e.g. ChatGPT and Claude).
- A neural network must be trained, whereby the weights are iteratively nudged to reduce the prediction error on the provided data.

Key caveat: the fitted network is a *black box* – it predicts well, but cannot explain why. We will come back to this.

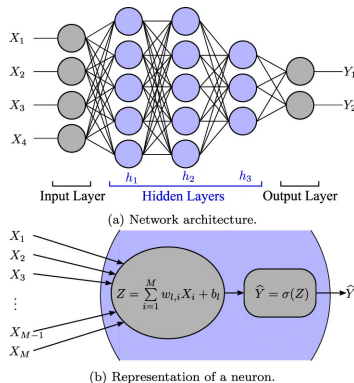


Figure 2: A feed-forward neural network (Ihme et al., 2022).

The surrogate idea

Replace an expensive computation (a simulator, an optimiser, ... or a *laboratory experiment*) with a cheap ML model trained on its input–output behaviour.

- Well established in chemical engineering (Henao & Maravelias, 2011; Granacher et al., 2021).
- Already used for LCA: ML surrogates for expensive life cycle inventory simulations (Aboagye et al., 2025).
- Once trained, evaluation is essentially instant $\Rightarrow$ real-time what-if analysis and visualisation.

Why uncertainty matters

- Neural networks are notoriously **overconfident** (Guo et al., 2017).
- In a high-stakes setting (e.g. involving toxic chemicals and industrial cost) a system must never prove a confident wrong answer.
- A good model should report its confidence in a prediction, and its confidence should be reliable.
- One idea is to, instead of predicting a value directly, predict a probability distribution $\mathcal{N}(\mu, \sigma^2)$ over the space of possible output values.
- The predicted variance σ^2 represents the uncertainty.

Two kinds of uncertainty

Aleatoric = “noise”

- Inherent randomness in the data itself
- **Irreducible**: more data will not help

Epistemic = “ignorance”

- The model simply hasn't seen enough data in that region
- **Reducible**: more data *does* help

Derived from the law of total variance:

$$\text{Var}(y \mid \theta) = E(\text{Var}(y \mid \theta)) + \text{Var}(E(y \mid \theta)) = \underbrace{E(\sigma^2)}_{\text{aleatoric uncertainty}} + \underbrace{\text{Var}(\mu)}_{\text{epistemic uncertainty}}$$

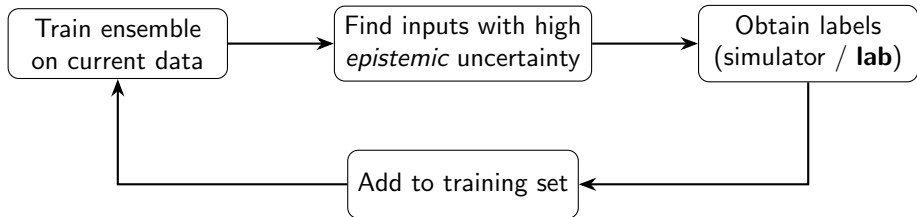
Deep ensembles: uncertainty from disagreement

- Train M copies of the same network, each starting from different random weights (here: $M = 5$). (Lakshminarayanan et al., 2017)
- Each network also predicts its own error bar (a mean *and* a variance) – like reporting a value $\pm$ standard deviation.

Reading the ensemble

- Models **agree**, small individual error bars $\Rightarrow$ confident prediction
- Models **disagree** $\Rightarrow$ *epistemic* uncertainty: the model needs more data here
- Models agree but all report **wide** error bars $\Rightarrow$ *aleatoric*: the problem itself is noisy

Active learning: let the model order its own experiments

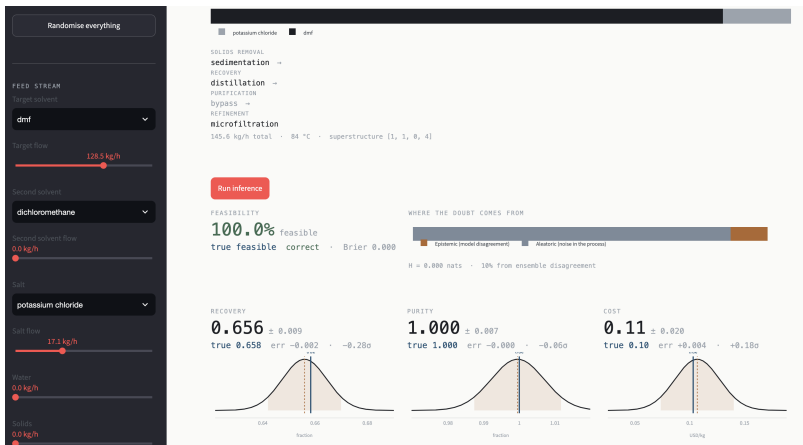


- Data is acquired where the model is most ignorant – the most informative experiments first.
- **The idea:** replace the simulator with a *laboratory*, and the model specifies the data it still requires, based on the data it's uncertain about.
- Granacher et al. (2021) achieve good results with this approach.

The pipeline at a glance

- 1 **Data generation:** reimplemented the equations of Chea et al. as an “oracle”: give it any waste stream and any pathway, it returns feasibility, recovery, purity, cost.
- 2 **Model:** one neural network ensemble ($M = 5$) with four output heads – feasibility (yes/no), recovery fraction, purity fraction, cost per kg – each predicting a probability distribution over the possible values.
- 3 **Uncertainty:** decomposed into aleatoric + epistemic.
- 4 **Active learning:** epistemic uncertainty guides data acquisition.
- 5 **Browser visualiser** for interactive exploration.

A simple visualiser



Inputs (24 numbers)

- Waste stream: two solvents + water, salt, solids
- Physical properties: relative volatilities, boiling points, densities, heats of vaporisation, ...
- Temperature
- The chosen pathway (which unit operation at each of the four stages)

Outputs (each with uncertainty)

- Is the pathway feasible at all?
- Fraction of solvent recovered
- Purity of the recovered solvent
- Cost per kg recovered

Initial training data

- 10^6 simulated flowsheet evaluations

How well does it predict?

Target	Metric	Value
Feasibility	Accuracy	0.999
Recovery	RMSE	0.012
Purity	RMSE	0.019
Cost per kg	RMSE	0.183

Table 1: Performance on unseen test data.

- Recovery and purity are fractions in $[0, 1]$: an RMSE of ~ 0.02 – 0.03 means the model is typically off by 2–3 percentage points.
- Evaluation is effectively instantaneous, allowing for interactive use.

Are the error bars honest? Calibration

- **Calibration:** when the model says “90% confidence”, it should be right 90% of the time.
- A calibration plot compares claimed vs. observed confidence: the ideal is the diagonal.
- My raw ensembles were *underconfident* – error bars too wide. A simple global rescaling of the predicted variance fixes this.

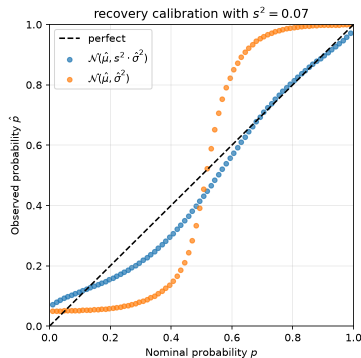


Figure 3: Calibration for the recovery prediction, before and after variance scaling.

Does active learning beat random data collection?

- No, not in my experiments.
- Probable cause: *sampling bias* – always chasing the most uncertain points breaks the i.i.d. assumption. Mixing in random samples helps, but doesn't fully cure it.
- The theoretical *framework* is still solid: a model that can point to the experiments it needs could produce interesting research. Hand-crafted data is expensive, its scope thus of importance.
- Future work should try to counteract sampling bias.

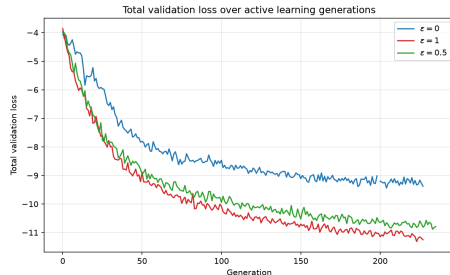


Figure 4: Validation loss over active learning generations.

The pure-solvent test

Feed the model a stream of **100% pure target solvent** and a pathway that **bypasses every single unit operation**. What is the purity of the output?

- Any human: obviously 100%.
- The model: consistently *underestimates* the purity.
- Machine learning approximates statistical patterns, not causal reasoning – a known, fundamental limitation of deep learning (Marcus, 2018).
- Domain-specific evaluation is thus vital.

The case for interpretability

- A neural network predicts *what*, and in this case even *how sure* – but never *why*.
- Rudin (2018): for high-stakes decisions, use inherently interpretable models; post-hoc explanations of black boxes can be plausible but wrong.

A cautionary tale from medicine (Caruana et al., 2015)

A model learned that **asthma patients survive pneumonia more often**. True in the data – because they receive more aggressive care. An interpretable model exposed this rule so doctors could remove it. A black box would have silently applied it.

- For low-dimensional tabular data like ours, interpretable models (generalised additive models, EBMs) often match deep learning performance (Doohan et al., 2026) – the black box may be an unnecessary cost.

Generalised Additive Models (GAMs)

- GAMs fit models without high-order feature interaction.
- This makes their learned function interpretable (Lou et al., 2012).
- Doohan et al. (2026) argue that for low-dimensional tabular data deep learning offers no performance benefit.
- Refinements have been developed by Lou et al. (2013) and Nori et al. (2019).

$$g(y) = \beta + \sum_i f_i(x_i)$$

What remains

Takeaways for life science

- 1 **Surrogates work.** If you have an expensive simulation, an ML surrogate can make it interactive.
- 2 **Quantify uncertainty.** Never accept a bare point prediction from an ML model in a scientific context. Prediction uncertainty must be embedded in the architecture and calibrated. Ensembles are a simple, robust way to achieve this end (Lakshminarayanan et al., 2016).
- 3 **Active learning fits the lab.** A model that requests its most informative next experiment could prove to be a highly useful tool.
- 4 **Prefer interpretable models** for tabular scientific data – the accuracy trade-off is marginal and you can work with rules that domain experts can check and correct.
- 5 **Always probe edge cases.** Machine learning never works like human brain. It's always a statistical approximation, regardless of how clever it seems.

- Built an uncertainty-aware neural network surrogate for solvent recovery flowsheet evaluation: 99.8% feasibility accuracy, recovery/purity typically within 2–3 percentage points.
- Uncertainty is decomposed into reducible (epistemic) and irreducible (aleatoric) parts and is well calibrated after a simple correction.
- Active learning should work in theory; acquisition strategies need refinement.
- Next step: an *interpretable* model (i.e., a GAM or some variation) built jointly with domain experts – and eventually, real laboratory data in the loop.

Thank you! Questions?

Code: `github.com/maxjappert/ml-superstructure-solvent-recovery`